6GGW / NetSwitch — functions, scientific names, F.A.T. & safety grouping

For each computing function in the product: its **official / scientific name**, the **approach** (what established method it is), the **Factory Acceptance Test (F.A.T.)** that proves it works, and a **safety / ethics group**. This is the pack you hand to a reviewer or an acceptance auditor. Numbering matches `MATH-INDEX.md` (M1–M40).

A note on “147 versions of AI” and STAC

I want to be straight with you rather than pad this out: the product does **not** contain 147 AI models, and there isn’t a real, citable list of “147 versions of AI” I can put my name to. What the system *does* contain is a set of **deterministic, auditable numerical and signal-processing functions** plus **one deterministic system-approval check** (the `approve` binary — your “the server’s AI is the same as the thin client’s”): same binary, same input, same answer everywhere, no black-box model. That is a genuine strength for F.A.T. and ethics — every output is reproducible and explainable.

- **STAC** (Securities Technology Analysis Center) and **MLPerf** are benchmark *bodies/suites*, not AI versions. If the “147” comes from a catalogue like that, send me the source and I’ll map our functions to whichever of its tests apply.
- If instead you want the classical-AI *methods* the system leans on named properly, the honest list is short and real: DCT/DSP transforms, entropy coding, numerical ODE integration, hashing, companding — all below. I’d rather give you 40 real, testable functions than 147 invented labels.

Safety / ethics groups

Group	Meaning	Care level
G1 Deterministic-compute	pure maths, reproducible, no learned model, fully auditable	low risk
G2 Signal/media	DSP on media frames; lossy but measured (PSNR/entropy)	low risk
G3 Measurement/telemetry	reads local hardware counters; no personal data	low risk
G4 Identity/security	handles secrets/keys; single-use, salted, never logged plaintext	**high care**
G5 Data plane	stores device/session rows in SQL; network params only	medium care

Ethics emphasis: no user tracking, no biometric/behavioural profiling, no covert data collection. Telemetry is machine-local; identity is one-time codes and TLS.

Functions — name · approach · F.A.T. · group

#	Function	Official / scientific name	Approach	F.A.T. (how proven)	Group
M1–M3	DRAMM kernel / workload	iterated fixed-point map (closed form)	precomputed Σ1/√k, deterministic iterate	result `0.120493199688742`, hash `1b8ab116…`	G1

AI2ORBIT CO. Network system Pat. pend. Founded in 2018. Copyright 2018-2027. All rights reserved.

				identical on Linux+Windows	
M4	result_hash	MurmurHash3 finalizer (fmix64)	avalanche bit-mix of IEEE-754 bits	same bits ⇒ same hash, verified cross-platform	G1
M5	approval_ratio	symmetric mean ratio √((a/b+b/a)/2)	quasi-arithmetic mean, =1 iff equal	1.000 on match, >1 on skew; 3× skew rejected	G1
M6/M7	DCT-II / IDCT-III	Discrete Cosine Transform (Ahmed 1974)	JPEG/H.26x separable 8×8 transform	Q=1 round-trip ≥ 58 dB PSNR (near-lossless)	G2
M8	powered/unpowered split	coefficient energy thresholding		C	≥ 1% peak ⇒ significant
M9	entropy	Shannon entropy (1948), bits & nats	−Σ p·log p over quantised magnitudes	matches hand calc on known histogram	G2
M10	PSNR / requant	Peak Signal-to-Noise Ratio	10·log10(range²/MSE)	monotone with Q; 58 dB at Q=1	G2
M11	best-scaler	rate–distortion knee	argmax dB per bit above floor	picks the elbow of the R/D curve	G2
M12/M26	spectral efficiency	Shannon–Hartley banding	bits/s per Hz × bandwidth	throughput scales linearly with MHz	G2
M13/M14	ddtm_constant(_calc)	log-domain closed form	LOG10(SQRT(…)/120W)	reproduces your −0.090908889 once T1/T2/LLR given	G1
M15	hyper_downscale	linear scaling	c/f	exact ratio, unit-tested in SQL	G1
M16	thrice_sinwave	3-term Fourier sum		k	·Σ a_i·sin(2π f_i t)
M17/M18	lnlog / binary_lnlog	iterated logarithm ln(ln x)	double-log domain	lnlog(e^e)=1 exactly	G1
M20/M21	μ-law / RTP	G.711 companding + RFC 3550	8-band logarithmic PCM, RTP packing	25/25 packets relayed intact both OS	G2
M22/M23	PSNR / blockiness	Wang–Bovik blockiness index	no-reference 8×8 grid pixelation (dB)	rises with real block artefacts	G2
M24/M25	battery ODE	Runge–Kutta–Fehlberg 4(5) + Laplace	numerical + analytic ODE, cross-checked	RKF45 vs analytic agree ~1e-15	G1
M27	split/rejoin	byte-exact partition	cut at fixed %s, checksum rejoin	SHA of rejoin == original	G1
M29/M30	ops/s, FPS, MFLOPS	LINPACK/DGEMM + FMA throughput	FLOP-counted wall-clock timing	checksummed so work isn't elided	G3
M31/M32	RDTSC MHz / radio reduce	calibrated time-stamp counter	cycle count ÷ calibrated interval	MHz stable across runs	G3
M33	IOPS/MB-s/latency	fsync-honest disk benchmark	seq/random with real fsync	percentiles reproduce on same disk	G3
M34	thermal headroom	trip-margin monitor	per-part temp vs trip point	flags throttle swing	G3
M35/M36	rerouting / distance	least-cost path + RTT	cost=RTT(1+load), km	least-cost POP chosen;	G3

AI2ORBIT CO. Network system Pat. pend. Founded in 2018. Copyright 2018-2027. All rights reserved.

		geodistance	from handshake	km ∝ RTT	
M37	SHA-256	FIPS 180-4	standard secure hash	empty-string digest = `e3b0c442…b855`	G4
M38	backup-code gen	Crockford base32, CSPRNG	50-bit entropy, unbiased byte%32	500 unique codes, no I/L/O/U	G4
M39	prefix_to_mask	CIDR → dotted mask	bit arithmetic	/24 → 255.255.255.0	G5
M40	TLS handshake	TLS 1.3 (EC-256/RSA)	standard key exchange + cert verify	negotiates + verifies cert	G4

F.A.T. procedure (how to run acceptance)

1. Build every module on the target OS (`build-all.sh` / MinGW).
2. Run `ggw_report` — capture MFLOPS + compatibility verdict (attach the HTML/PDF).
3. Run each module’s self-test / known-answer check in the table above.
4. Confirm the cross-platform hashes match (approve `1b8ab116…`, SHA-256 empty digest).
5. File the report + this table as the acceptance record.

Every function here is deterministic or measured — there is no hidden model whose output you can’t reproduce. That is deliberate, and it is what makes the system pass a real Factory Acceptance Test.

AI2ORBIT CO. Network system Pat. pend. Founded in 2018. Copyright 2018-2027. All rights reserved.